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Blessed G/Michael Catholic General Secondary School

Mathematics Test one 2018 E. C for grade 12th

Name: Hanna Ayalew Section: 12th C No:-24. **Time allowed 1:30**

I. Choose the best answer among the given alternatives.

C 1. Which one of the following is the 4th term of the sequence $a_{n+2} = na_1 + (n+1)a_2$, for $n \geq 2$ and $a_1 = -1, a_2 = 2$

A. 6 B. 5 C. 4 D. 7

C 2. Which one of the following is an arithmetic sequences?

A. $a_n \left(\frac{1}{2}\right)^n$ B. an $= (-1)^0$ C. an $= 4$ D. an $= 1+(-1)^n$

A 3. Give an arithmetic sequence with the fourth term is 8 and the 8th term is 10, the first term of the sequence is ----------.

A. $13/2$ B. $16/3$ C. $15/2$ D. 14

B 4. The total number of whole numbers that are less than 1000 and divisible by 7 is ---------.

A. 40 B. 143 C. 120 D. 30

A 5. Which one of the following sequence is geometric sequence

A. $gn = \left(\frac{3}{2}\right)^n$ B. $gn = (-1)^n$ C. gn = 4-2n D. gn = 0

B 6. The sum of whole numbers that are less than 100 and leave remainder 2 when divided by 5 is ----------.

A. 400 B. 990 C. 690 D. 390

A 7. $\sum_{k=1}^{\infty} \left(\frac{2^k+3^k}{6^k}\right) 2 =$ ----------

A. 3 B. 4/3 C. $\frac{103}{48}$ D. $\frac{40}{103}$

B 8. The sum of all two – digit whole numbers which are divisible by 11 is ----------.

A. 390 B. 495 C. 345 D. 290

C 9. When n – arithmetic means are inserted between 10 and 50, the sum of the resulting term is 240. How many terms are there?

A. 10 B. 9 C. 6 D. 14

B 10. Which one of the following is true for the sum of 8,11,14,-,-,-,-98

A. An = 2n+3 C. The common difference is 4

B. The number of terms is 31 D. The sequence is geometric

C 11. The geometric mean between $\sqrt{5} - \sqrt{2}$ and $\sqrt{5} + \sqrt{2}$ is ----------

A. 4 B. 3 C. $\sqrt{3}$ D. $\sqrt{5}$

A 12. The sum of $\left(\frac{1}{2}\right)^n$ for n – tends infinity is ----------

A. 1 B. ∞ C. 0 D. $\frac{1}{2}$

B 13. $\sum_{n=1}^{\infty} -(2/3)^n (-1)^n$ is ----------

A. 4 B. $2/5$ C. $2/3$ D. $4/5$

A 14. Which one of the following is convergent sequence

A. $(-1)^n$ B. 2^n C. $\left(\frac{5}{4^n}\right)$ D. $\left(\frac{4}{3}\right)^n$

15. The sum of the first n – term of the sequence $a_n = \frac{1}{n^2+n}$ is .....

A. $\frac{n}{n+1}$ B. $\frac{2n}{n^2+1}$ C. $\frac{4n}{2n+4}$ D. $\frac{n^2}{n^2+1}$

16. The sum of the first three terms of a geometric sequence is 8, and the sum from 4th to 6th terms is 512, the common ratio of the sequence is .....

A. 6 B. 2 C. 4 D. 1

17. The sequence $\{a_n\}$, $n \geq 1$ where $a_1 = 2$ *and* $a_{n+1} = an+4$ then $a_3+a_4+a_5+\cdots+a_{35}$ is equal to

A. 2460 B. 2458 C. 2442 D. 2450

18. The product of $3, 3^{1/3}, 3^{\frac{1}{6}}, 3^{\frac{1}{17}}, \ldots$ is .....,

A. $3\sqrt{3}$ B. $2\sqrt{3}$ C. $\sqrt{45}$ D. $\sqrt{6}$

19. The fraction form of $2.\overline{3}$ is .....

A. $4/5$ B. $\frac{21}{9}$ C. $3/4$ D. $6/5$

20. The two numbers between 3 and 81 so that the resulting sequence is geometric progression is .....

A. 3, 9 B. 3, 6 C. 9, 27 D. 27, 51

Short answer

21. A contest offers 20 prizes with a total value of birr 20,000 If the difference in the value between consecutive prize is birr 400, what is the value of the first prize? 1800

22. If the nth partial sum of an arithmetic sequence is $4n^2$, then the nth term of the sequence is $a_n = 8n-4$

23. Aman saved birr 1600 in 8 years. In each year after the first he saved birr 100 more than he did in the preceding year. How much did he saved in the first year. 150

24. $\sum_{n=0}^{\infty} \cos(n\pi)$ = ...0.....?

25. A ball is dropped from height of 20 meter. Each time it drops h meter, It rebounds $\left(\frac{1}{2}\right)$ meters, the total vertical distance traveled by the ball is .....60m.....

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